

Tyrone Marhguy

Computer Engineering — Software Engineering / Systems / Backend / Developer Tools

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SUMMARY

Computer Engineering junior building systems from search/indexing and transactional storage to compilers, developer tools, and production APIs. Shipped public software and merged C++/Python fixes into major open-source EDA codebases; strongest in Rust/C++, backend systems, performance, and correctness-focused engineering.

EDUCATION

University of Pennsylvania
B.S.E. Computer Engineering

Philadelphia, PA
Aug. 2024 – May 2028

EXPERIENCE

Software Engineer Intern
Aragorn AI

May 2026 – Aug. 2026
Remote

- Shipped **production backend services and REST APIs** for AI integrations; resolved cross-service failures blocking releases and hardened service boundaries through architecture and code reviews.

Open-Source Software Contributor

2026 – Present

OpenROAD, Verilator, LibreLane, OpenFPGA

C++ / Python / EDA Infrastructure

- Merged upstream fixes into production engineering tools: repaired an **OpenROAD LEF58/ODB parser**, corrected **Verilator Linux peak-memory accounting**, and fixed LibreLane/Yosys synthesis compatibility and reporting; LibreLane fixes shipped in releases **3.0.8** and **3.0.10**.

Hardware & Firmware Engineer
Vero Electric

May 2026 – Aug. 2026
Philadelphia, PA

- Developed **embedded C/C++ fault-detection firmware** for a BQ79616 battery-monitoring system and debugged firmware/hardware integration on a **240 kWh / 120 kW** platform using lab instrumentation.

SELECTED SOFTWARE & SYSTEMS PROJECTS

- SeaLion — Full-Text Search Engine** | Rust / BM25 / Search / Storage GitHub
 - Built a full-text search engine from first principles with **Unicode analysis**, **BM25-ranked phrase search**, **persistent .seal segments**, **index generations**, and **segment merging**; validated optimized behavior against a reference oracle with **89 passing tests**.
- FigDB — Transactional Storage Engine** | Rust / LSM / WAL / Tokio GitHub
 - Implemented ordered KV storage, a **checksummed WAL**, WAL-backed memtables, SSTables, LSM flush/merge, manifests, and compaction from first principles; maintained **59 passing tests** across persistence and recovery behavior.
- Lobster — Systems Language & Optimizing Compiler** | Rust / Compilers / Tomato32 GitHub
 - Designed a statically typed systems language and compiler targeting conventional and dual-LUT3/Tomato architectures; implemented a lexer, **Pratt parser**, name resolution, and type checking behind **lobster check** across **nine Rust crates**.
- FramePort — HDMI Capture Inside VS Code** | TypeScript / VS Code API / FFmpeg Open VSX | GitHub
 - Published a VS Code extension for live HDMI/USB capture, pixel inspection, PNG screenshots, and H.264 recording; reached **1,250+ Open VSX downloads** and implemented backpressure, lifecycle recovery, save safety, diagnostics, and encode/decode regression coverage.
- Envelop — Browser-to-FPGA Remote Compute** | TypeScript / Swift / Kotlin / BLE Website | GitHub
 - Built a browser-to-hardware messaging system connecting remote clients to physical Tomato FPGA computers through verified native BLE bridges; implemented a durable job queue, bridge leases, replay-safe handling, binary **ENVELOP/1** protocol, and web, Swift, and Android clients.

TECHNICAL SKILLS

Languages: Rust, Python, C/C++, TypeScript/JavaScript, Swift, Java/Kotlin, Bash, SQL, Embedded C

Systems / Backend: Linux, concurrency, networking, storage engines, search/indexing, compilers, FastAPI, Node.js, REST APIs, PostgreSQL, Supabase, Redis, binary protocols

Tooling / DevOps: Docker, Kubernetes, GitHub Actions, CI/CD, Nginx, FFmpeg, VS Code Extension API, pytest, browser/integration testing